1 Probability Axioms & Rules

$P(S) = 1$, $P(\emptyset) = 0$, $0 \le P(A) \le 1$
$P(A) = P(A \cap B) + P(A \cap B^c)$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$
Disjoint ($A \cap B = \emptyset$): $P(A \cup B) = P(A) + P(B)$
Classical: $P(E) = \#$ desired/$\#$ total

1.1 Conditional Probability

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \quad 0 \le P(E|F) \le 1, \quad P(S|F) = 1$$

$P(E \cap F) = P(E|F)P(F) = P(F|E)P(E)$
Use when: given that something already happened, find new P.

1.2 Chain Rule

$P(A_1 A_2 \cdots A_n) = P(A_1)\, P(A_2|A_1)\, P(A_3|A_1A_2) \cdots P(A_n|A_1 \cdots A_{n-1})$
Use when: things happening **in sequence** (draw w/o replacement, etc.)

1.3 Law of Total Probability

$P(B) = \sum_i P(B|A_i)P(A_i)$ (A_i partition S)
Two-event: $P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$
Use when: want $P(B)$ but it depends on **which scenario** happened (coin A vs B, defective vs good, etc.). Split into cases.

1.4 Bayes' Formula

$$P(A_i|B) = \frac{P(B|A_i)\, P(A_i)}{\sum_j P(B|A_j)\, P(A_j)}$$

Use when: you observe a **result** and want to find which **cause** is most likely. "Flips" $P(B|A) \to P(A|B)$.

2 Independence

$E \perp F \iff P(E \cap F) = P(E)\, P(F) \iff P(E|F) = P(E)$
If $E \perp F$: then $E \perp F^c$, $E^c \perp F$, $E^c \perp F^c$.
Disjoint events are **NOT** independent (unless one has $P = 0$).
A, B, C mutually indep. requires **all pairwise** indep. **AND** $P(ABC) = P(A)P(B)P(C)$.
$P(ABC) = P(A)P(B)P(C)$ alone does **not** imply pairwise indep.
Conditional indep.: $A \perp B|C$: $P(A \cap B|C) = P(A|C)\, P(B|C)$
$\implies P(A|BC) = P(A|C)$ (given C, knowing B gives no info on A)
Attraction/Repulsion:
$P(AB) < P(A)P(B)$: repulsion $P(AB) = P(A)P(B)$: indep.
$P(AB) > P(A)P(B)$: attraction (disjoint = extreme repulsion)
If $A \perp B$: $P(A \cap B^c) = P(A)(1 - P(B))$ (useful shortcut)
Shared elements shortcut (equally likely outcomes only):
$|E \cap X| = |X| \cdot P(E)$; if this holds, $E \perp X$.

3 Geometric Series

Infinite: $\sum_{n=0}^{\infty} ar^n = \dfrac{a}{1-r}$, $|r| < 1$ Finite: $\sum_{n=0}^{N-1} r^n = \dfrac{1 - r^N}{1 - r}$
Use when: summing $(1-p)^k$ terms, e.g. geometric PMF sums.

4 Random Variables & CDF

$X : S \to \mathbb{R}$. $F_X(u) = P(X \le u)$
$0 \le F \le 1$; $F(-\infty) = 0$; $F(\infty) = 1$; non-decreasing; right-continuous
$P(X > u) = 1 - F(u)$ $P(X = u) = F(u) - F(u^-)$
$P(a < X \le b) = F(b) - F(a)$ $P(a \le X \le b) = F(b) - F(a^-)$
If F continuous at a: $P(X = a) = 0$, all interval forms equal.

4.1 Discrete R.V. – PMF

Staircase CDF; $P(X = a) > 0$; $p_X(u) = P(X = u)$; $\sum_u p(u) = 1$

4.2 Continuous R.V. – PDF

Smooth CDF; $P(X = a) = 0$; $f_X(u) = dF_X/du$; $\int f(u)\, du = 1$
$P(a \le X \le b) = \int_a^b f_X(u)\, du$; $F_X(u) = \int_{-\infty}^u f_X(z)\, dz$
PDF can be > 1 (area is P, not value).
PDF on (a, b) only $\Rightarrow$ CDF has 3 pieces (0, ramp, 1).

5 Distributions – Formulas & Recognition

5.1 Bernoulli: $X \sim \mathbf{Ber}(p)$

$P(X = 1) = p$, $P(X = 0) = 1 - p$. $E[X] = p$, $\text{Var} = p(1 - p)$.
Scaled: $W = (b-a)X + a$: $E = (b-a)p + a$, $\text{Var} = (b-a)^2 p(1-p)$
Recognize: single trial, two outcomes, PMF has 2 spikes. R.v. takes only 2 distinct values.

5.2 Binomial: $X \sim \mathbf{Bin}(n, p)$

$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k = 0, \ldots, n$; $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
$E[X] = np$, $\text{Var}(X) = np(1 - p)$
$X \sim \text{Bin}(n_1, p)$, $Y \sim \text{Bin}(n_2, p)$ indep.: $X + Y \sim \text{Bin}(n_1 + n_2, p)$
Recognize: (1) **fixed** $\#$ of trials, (2) trials **independent**, (3) **two** possible outcomes, (4) **constant** p. Asks: "how many successes?"

5.3 Poisson: $X \sim \mathbf{Pois}(\lambda)$

$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$ $E[X] = \lambda$, $\text{Var}(X) = \lambda$
Recognize: "how many events in a fixed time/space?" λ = average rate. E.g. "how many texts in the next hour?" ($\lambda = 4$/hr).

5.4 Geometric: $X \sim \mathbf{Geo}(p)$

$P(X = k) = p(1 - p)^{k-1}$, $k = 1, 2, \ldots$ $P(X > k) = (1 - p)^k$
$F_X(u) = 1 - (1 - p)^{\lfloor u \rfloor}$; $E[X] = 1/p$, $\text{Var} = (1 - p)/p^2$
Recognize: "how many trials until the **first** success?" Indep. trials, constant p, waiting for 1st success. **Memoryless** (discrete).

5.5 Negative Binomial

$P(X = n) = \binom{n-1}{r-1} p^r (1 - p)^{n-r}$, $n = r, r+1, \ldots$
Recognize: "how many trials to get r successes?" (Geometric is $r = 1$ case.)

5.6 Uniform Continuous: $X \sim U(a, b)$

$f_X(u) = \frac{1}{b-a}$, $a \le u \le b$; $F_X(u) = \frac{u-a}{b-a}$
$E[X] = \frac{a+b}{2}$, $\text{Var} = \frac{(b-a)^2}{12}$
$P(c < X < d) = \frac{d-c}{b-a}$; on $U(0, 1)$: $P(c < X < d) = d - c$
Recognize: all values equally likely in a range; flat PDF.

5.7 Gaussian (Normal): $X \sim N(m, \sigma^2)$

$f_X(u) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-(u-m)^2/(2\sigma^2)}$ $E[X] = m$, $\text{Var} = \sigma^2$
Standard: $Z \sim N(0, 1)$, CDF $= \Phi(z)$, $\Phi(-a) = 1 - \Phi(a)$
Convert: $F_X(u) = \Phi\left(\frac{u-m}{\sigma}\right)$
Linear transform: $Y = aX + b \implies Y \sim N(am + b,\ a^2\sigma^2)$
Recognize: bell-shaped; linear combo of Gaussians is Gaussian.
Useful Φ values: $\Phi(1) \approx 0.841$, $\Phi(2) \approx 0.977$, $\Phi(3) \approx 0.999$
68-95-99.7 rule: $P(|X - m| < k\sigma)$ for $k = 1, 2, 3$.
Mean from quadratic exponent: if exponent is $-ax^2 + bx$, mean $= b/(2a)$

5.8 Exponential: $X \sim \mathbf{Exp}(\lambda)$

$f_X(u) = \lambda e^{-\lambda u}$, $u \ge 0$; $F_X(u) = 1 - e^{-\lambda u}$
Survival: $P(X > u) = e^{-\lambda u}$; $E[X] = 1/\lambda$, $\text{Var} = 1/\lambda^2$
Memoryless: $P(X > u + v \mid X > u) = P(X > v)$
Min: $Y = \min(X_1, \ldots, X_n)$, $X_i \sim \text{Exp}(\lambda_i)$ indep. $\Rightarrow Y \sim \text{Exp}(\sum \lambda_i)$
$P(X < Y) = \lambda_X/(\lambda_X + \lambda_Y)$ for indep. exponentials.
Recognize: waiting time until event; "how long until…?"; continuous memoryless. PDF is decaying exponential.

5.9 Kernel Matching

Use when: problem guarantees a valid PDF; match the kernel to a known distribution.
Domain clues: $-\infty < x < \infty$: Normal; $0 < x < 1$: Beta; $x > 0$: Exp/Gamma.
Completing-the-square trap: $\propto e^{-ax^2+bx}$ is Normal—complete the square to get $e^{-a(x-b/2a)^2}$, read off mean $= b/(2a)$, $\sigma^2 = 1/(2a)$.

6 Expected Value

Discrete: $E[X] = \sum u\, p_X(u)$ Continuous: $E[X] = \int u\, f_X(u)\, du$
$E[g(X)] = \sum g(u)\, p_X(u)$ or $\int g(u)\, f_X(u)\, du$
$E[g(X,Y)] = \sum\sum g(u,v)\, p_{X,Y}$ or $\iint g(u,v)\, f_{X,Y}\, du\, dv$
Linearity: $E[aX + b] = aE[X] + b$; $E[X + Y] = E[X] + E[Y]$ **always**
Bounded r.v.: $E[X] = \int_0^{\infty}(1 - F(u))\, du - \int_{-\infty}^0 F(u)\, du$
Indicator ($I \in \{0, 1\}$): $E[I] = P(\text{event})$. Break complex r.v. into sum of indicators.
Use indicators when: counting something (e.g. $\#$ of matches, $\#$ correct).
Step-fn PDF: $E[X] = \sum(\text{Area}_i \cdot \text{midpoint}_i)$
Use centroid when: uniform joint PDF or piecewise-constant PDF.

7 Moments & Variance

$\text{Var}(X) = E[X^2] - (E[X])^2$ $\sigma_X = \sqrt{\text{Var}(X)}$
$\text{Var}(a + bX) = b^2\text{Var}(X)$ $\text{Var}(X + c) = \text{Var}(X)$
$E[X^2] = \text{Var}(X) + (E[X])^2$

8 Functions of a R.V.

Discrete: $P(Y = v) = \sum_{\{x: g(x) = v\}} P(X = x)$
Change of var. (monotonic): $f_Y(y) = f_X(g^{-1}(y))\, |dg^{-1}/dy|$
Use when: $Y = g(X)$ with g **strictly** increasing or decreasing.
Linear: $Y = aX + b$: $f_Y(u) = \frac{1}{|a|} f_X\left(\frac{u-b}{a}\right)$
CDF method (general):
Use when: g is **not monotonic**, or you're unsure. Always works.
1) $F_Y(u) = P(g(X) \le u) \to$ solve for $X \to$ express via $F_X \to$ differentiate.

8.1 Leibniz Rule

$\frac{d}{du}\int_{h(u)}^{g(u)} f(z)\, dz = f(g(u))\, g'(u) - f(h(u))\, h'(u)$
Use when: differentiating a CDF that has variable limits (CDF method step).

9 Joint Distributions

Joint CDF: $F_{X,Y}(u, v) = P(X \le u,\ Y \le v)$
Marginals: $F_X(u) = F_{X,Y}(u, \infty)$; $F_Y(v) = F_{X,Y}(\infty, v)$
Joint $\to$ marginals loses info (can't recover joint from marginals alone).
Discrete: $p_{X,Y}(u, v) = P(X = u, Y = v)$.
Marginal: $p_X(u) = \sum_v p_{X,Y}(u, v)$ ("sum out the other").
Continuous: $P((X, Y) \in T) = \iint_T f_{X,Y}\, du\, dv$; $\iint f_{X,Y}\, du\, dv = 1$
$f_{X,Y} = \frac{\partial^2 F_{X,Y}}{\partial u\, \partial v}$; Marginal: $f_X(u) = \int f_{X,Y}(u, v)\, dv$ ("integrate out the other").
Uniform joint: $f = 1/\text{Area}(T)$ on region T.
P = valid area/total area; marginal PDF = (slice length) / Area.
Recognize: "point chosen uniformly at random in a region."

9.1 Independence of R.V.s

$X \perp Y \iff f_{X,Y} = f_X f_Y$ (or PMF/CDF version).
Test: if $f_{X,Y}(u, v) = g(u) \cdot h(v)$ over a **rectangular** domain, indep.
If $X \perp Y$: $E[XY] = E[X]E[Y]$; $\text{Cov} = 0$; $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

9.2 Conditional Distribution

$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$; $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$
Use when: given $Y=y$, find distribution of X.
If $X \perp Y$: $f_{X|Y}(x|y) = f_X(x)$ (conditioning gives no new info).

9.3 Conditional Expectation

$E[X|Y=y] = \sum x\, p_{X|Y}(x|y)$ or $\int x\, f_{X|Y}(x|y)\, dx$
Law of total expectation: $E[X] = E[E[X|Y]]$
Variance decomposition: $\mathrm{Var}(X) = E[\mathrm{Var}(X|Y)] + \mathrm{Var}(E[X|Y])$
Use when: $E[X]$ or $\mathrm{Var}(X)$ is hard directly; condition on easier Y.

10 Sum of R.V.s: $Z = X + Y$ (Convolution)

General: $f_Z(z) = \int f_{X,Y}(z-v,\, v)\, dv$
Indep.: $f_Z = (f_X * f_Y)(z) = \int f_X(z-v)\, f_Y(v)\, dv$
Use when: want PDF of a **sum** (or difference) of two r.v.s.

11 Max and Min of R.V.s

$W = \max(X_1, \ldots, X_n)$: $F_W(t) = \prod F_{X_i}(t)$ (indep.)
i.i.d.: $F_W = [F_X]^n$; $f_W = n[F_X]^{n-1} f_X$
$Z = \min(X_1, \ldots, X_n)$: $1 - F_Z(t) = \prod(1 - F_{X_i}(t))$ (indep.)
i.i.d.: $F_Z = 1 - [1 - F_X]^n$
n i.i.d. $U(0,1)$: Max $F = t^n$, $f = nt^{n-1}$; Min $F = 1-(1-t)^n$
Use when: "best of n" or "worst of n" or "first to finish/fail." Key idea: max $\le t \iff$ **all** $\le t$; min $> t \iff$ **all** $> t$.
$P(\max(X,Y) < u) = P(X < u \cap Y < u)$; $P(\min(X,Y) < u) = P(X < u \cup Y < u)$

12 Covariance & Correlation

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$ $\quad \mathrm{Cov}(X,X) = \mathrm{Var}(X)$
$\mathrm{Cov}(X,\, Y_1 + Y_2) = \mathrm{Cov}(X, Y_1) + \mathrm{Cov}(X, Y_2)$
$\mathrm{Cov}(aX,\, bY) = ab\,\mathrm{Cov}(X,Y)$ $\quad \mathrm{Cov}(X,\, aX+b) = a\,\mathrm{Var}(X)$
$\mathrm{Cov}(X+Y,\, W+Z) = \mathrm{Cov}(X,W) + \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,W) + \mathrm{Cov}(Y,Z)$
Correlation coefficient: $\rho_{XY} = \mathrm{Cov}(X,Y)/(\sigma_X \sigma_Y)$, $-1 \le \rho \le 1$
$\mathrm{corr}(X,\, aX+b) = \mathrm{sign}(a)$
Outcomes on $Y = aX + b$: $a > 0 \Rightarrow \rho = 1$; $a < 0 \Rightarrow \rho = -1$
$X \perp Y \Rightarrow$ uncorrelated (**converse false** in general).
Jointly Gaussian: uncorrelated $\iff$ independent.
If $\rho = 0$: joint PDF factors as $f_X \cdot f_Y \Rightarrow$ independent.

12.1 Variance of Sums

$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$
$\mathrm{Var}(a + bX + cY) = b^2\mathrm{Var}(X) + c^2\mathrm{Var}(Y) + 2bc\,\mathrm{Cov}(X,Y)$
If $X \perp Y$ or uncorrelated: $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$
n indep.: $\mathrm{Var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \mathrm{Var}(X_i)$
n i.i.d.: $\mathrm{Var}(\sum X_i) = n\,\mathrm{Var}(X)$; $\mathrm{Var}(\bar{X}) = \mathrm{Var}(X)/n$

13 Inequalities & Limit Theorems

13.1 Markov Inequality

$X \ge 0$: $\boxed{P(X \ge \alpha) \le \dfrac{E[X]}{\alpha}}$
Use when: bounding P for a **non-negative** r.v., only $E[X]$ known.
E.g. avg income \$50K: $P(X \ge 200\text{K}) \le 50/200 = 25\%$.

13.2 Chebyshev Inequality

$\boxed{P(|X - E[X]| \ge \varepsilon) \le \dfrac{\mathrm{Var}(X)}{\varepsilon^2}}$
Use when: bounding how far from the mean; need $E[X]$ and $\mathrm{Var}(X)$. Tighter than Markov. Works for **any** distribution.

13.3 Weak Law of Large Numbers

$\bar{Y}_n = \frac{1}{n}\sum X_i$; $E[\bar{Y}_n] = m$; $\mathrm{Var}(\bar{Y}_n) = \sigma^2/n$
$P(|\bar{Y}_n - m| \ge \varepsilon) \le \sigma^2/(n\varepsilon^2) \to 0$
Use when: asked about sample mean converging to true mean, or bounding error of empirical average.

13.4 Central Limit Theorem

$Y_n = \frac{\sum X_i - n\mu}{\sigma\sqrt{n}}$; $P(Y_n \le u) \to \Phi(u)$ as $n \to \infty$
Use when: approximating distribution of a **sum** of many i.i.d. r.v.s. Normalized sum $\to$ Gaussian regardless of original distribution.

14 Comparing Two R.V.s

$P(X < Y) + P(Y < X) + P(X = Y) = 1$
Discrete: $P(X < Y) = \sum_k P(X < k)\, P(Y = k)$
$P(X = Y) = \sum_k P(X = k)\, P(Y = k)$
Use when: asked "which is bigger?" or "P they are equal."

15 Transformations: $Z = XY$, $W = X/Y$

$f_{Z,W}(a,b) = \frac{1}{|b|}\, f_{X,Y}\left(\frac{a}{b},\, b\right)$
Use when: given joint of (X,Y), want joint of product and ratio.

16 Gaussian Integrals (All Variations)

$\int_{-\infty}^{\infty} e^{-at^2}\, dt = \sqrt{\frac{\pi}{a}}$ $\quad$ ($a > 0$, base case)
$\int_{-\infty}^{\infty} t^n\, e^{-at^2}\, dt = 0$ $\quad$ for **odd** n (by symmetry)
$\int_{-\infty}^{\infty} t^2\, e^{-at^2}\, dt = \frac{1}{2a}\sqrt{\frac{\pi}{a}} = \frac{\sqrt{\pi}}{2\,a^{3/2}}$
$\int_{-\infty}^{\infty} t^4\, e^{-at^2}\, dt = \frac{3}{4a^2}\sqrt{\frac{\pi}{a}} = \frac{3\sqrt{\pi}}{4\,a^{5/2}}$
$\int_0^{\infty} t\, e^{-at^2}\, dt = \frac{1}{2a}$ $\quad$ (half-line, odd integrand contributes)
$\int_0^{\infty} t^3\, e^{-at^2}\, dt = \frac{1}{2a^2}$
General even: $\int_{-\infty}^{\infty} t^{2k} e^{-at^2}\, dt = \frac{(2k)!}{k!\,(2a)^k} \cdot \frac{1}{2}\sqrt{\frac{\pi}{a}}$
Use when: computing $E[X^n]$ for Gaussian, or normalizing constants.
Key rule: odd powers over symmetric limits $(-\infty, \infty)$ always vanish.
To evaluate: match your integral to $\int t^n e^{-at^2} dt$ by identifying a.

17 Miscellaneous Formulas

Derangement: $D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$; $D_n = nD_{n-1} + (-1)^n$
Values: $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$, $D_6 = 265$
Use when: no item in its original position (e.g. hat problem).
Time reversal: $P(\text{unseen is target}) = K/N$ regardless of what you've seen.
Use when: drawing cards—prob. of next card doesn't depend on order.
Expected abs. diff: $E[|X - Y|] = 2\int_0^{\infty} F(t)(1 - F(t))\, dt$ (i.i.d.)
Use when: expected distance/gap/difference between two i.i.d. things.
Median: $F(A) = 0.5$; **Quadratic:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Log: $\ln(ab) = \ln a + \ln b$; $\ln(a/b) = \ln a - \ln b$; $\ln(a^b) = b \ln a$
Areas: $\triangle$: $\frac{1}{2}bh$; $\bigcirc$: πr^2; semicircle: $\frac{\pi r^2}{2}$; rectangle: lw
$e^{a+b} = e^a e^b$; $e^{\ln x} = x$; $\lim_{t\to\infty} e^{-t} = 0$; $\lim_{t\to\infty} te^{-t} = 0$; $e^{-\infty} = 0$; $e^{\infty} = \infty$
$n! = n \cdot (n-1) \cdots 2 \cdot 1$; $0! = 1$; $\binom{n}{0} = \binom{n}{n} = 1$
Complement trick: $P(\text{at least } 1) = 1 - P(\text{none})$.
Use when: "at least one" is hard; $P(\text{none})$ is often easier.
MMSE linear est.: $\hat{X} = aY + b$ minimizes $E[(X - \hat{X})^2]$:
$a = \rho_{XY}\,\frac{\sigma_X}{\sigma_Y}$, $\quad b = E[X] - a\,E[Y] \quad \Rightarrow \quad \hat{X} = E[X] + \rho\frac{\sigma_X}{\sigma_Y}(Y - E[Y])$
Use when: observe Y, want to estimate X linearly.
i.i.d.: independent **and** identically distributed (same PDF/PMF).
Poisson approx.: n large, p small, $\lambda = np$: $\mathrm{Bin}(n,p) \approx \mathrm{Pois}(\lambda)$.
Ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$; center (h,k), semi-axes a, b; $a > b$: wider; $b > a$: taller
Centroids: quarter-circle: $\bar{x} = \bar{y} = 4R/(3\pi)$; semicircle: $4R/(3\pi)$ on symmetry axis
Exponent rule: $a^n \cdot b^n = (ab)^n$ $\qquad |A| > B \iff (A > B) \cup (A < -B)$

17.1 Common Integrals

$\int e^{au}\, du = \frac{e^{au}}{a}$ $\qquad \int ue^{au}\, du = \frac{e^{au}}{a}\left(u - \frac{1}{a}\right)$
$\int_0^{\infty} u^n e^{-au}\, du = \frac{n!}{a^{n+1}}$ $\qquad \int \frac{du}{u} = \ln|u|$ $\qquad \int_0^1 u^n\, du = \frac{1}{n+1}$
$\int \sin(au)\, du = -\frac{\cos(au)}{a}$ $\qquad \int \cos(au)\, du = \frac{\sin(au)}{a}$
By parts: $\int u\, dv = uv - \int v\, du$ $\qquad \int \ln(x)\, dx = x\ln(x) - x$
$\int_a^b c\, du = c(b-a)$; $\int_a^b u\, du = \frac{b^2 - a^2}{2}$; $\int_a^b u^2\, du = \frac{b^3 - a^3}{3}$

17.2 When to Use What – Quick Guide

If you see...	Use...
"given that A happened"	Conditional P
sequence of dependent steps	Chain rule
want $P(B)$ across scenarios	Total probability
know result, want cause	Bayes' formula
find PDF of $Y = g(X)$	CDF method or CoV
sum/diff of two r.v.s	Convolution
best/worst of n r.v.s	Max/Min CDF
non-neg. r.v., bound P	Markov
bound deviation from mean	Chebyshev
avg of many samples $\to$ mean	WLLN
approx. dist. of large sum	CLT $\to \Phi$
count something complex	Indicator variables
uniform region, find P	Area ratio
$E[X]$ hard directly	Total expect.: $E[E[X\|Y]]$
observe Y, estimate X	MMSE linear estimator
"at least one" probability	Complement: $1 - P(\text{none})$
given Y, find dist. of X	Conditional PDF/PMF

17.3 Region Formulas for Joint PDFs

Full disk: $x^2 + y^2 \le r^2$, area $= \pi r^2$
Upper semicircle: $y = \sqrt{r^2 - x^2}$, area $= \pi r^2/2$
Right triangle w/ legs a, b: area $= ab/2$
Square $[0,a] \times [0,a]$: area $= a^2$

17.4 Common Pitfalls & Useful Tricks

$E[X^2] \ne (E[X])^2$ unless $\mathrm{Var}(X) = 0$.
$E[1/X] \ne 1/E[X]$ in general.
$E[XY] = E[X]E[Y]$ only if uncorrelated (not always!).
Uncorrelated $\not\Rightarrow$ independent (except jointly Gaussian).
Disjoint $\ne$ independent.
PDF > 1 is fine; it's not a probability (area is).
To find normalizing constant c: set $\int f(x)\, dx = 1$, solve for c.
Non-rectangular support region for joint PDF $\Rightarrow$ **not** independent.
Complete the square: $-ax^2 + bx = -a(x - b/2a)^2 + b^2/4a$
$\to$ use for Gaussian-like integrals that aren't in standard form.
$P(A \cup B) \le P(A) + P(B)$ (union bound—useful for "at most").
$e^x \approx 1 + x$ for small $|x|$; $(1 + x/n)^n \to e^x$ as $n \to \infty$.
Permutations: $P(n,k) = n!/(n-k)!$ (order matters).
Combinations: $\binom{n}{k} = n!/(k!(n-k)!)$ (order doesn't matter).
Sampling w/ replacement: n^k outcomes.
Sampling w/o replacement: $P(n,k)$ ordered, $\binom{n}{k}$ unordered.
$X \perp Y \Rightarrow E[g(X)h(Y)] = E[g(X)]\, E[h(Y)]$ for any g, h.
Mixed r.v.: CDF has both jumps (discrete) and smooth parts (continuous). P of continuous part + P of discrete spikes = 1.